A FINITE RING MAP IS OF FINITE TYPE, AND A FINITELY PRESENTED MODULE STRUCTURE GIVES FINITE PRESENTATION

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Allowed standard prerequisites (editorial).

- Module generators as algebra generators. Used in part (1) of the counted proof (lines 899-900). Shows finite as a module implies finite type as an algebra.
- Writing a presentation from module relations and structure constants. Used in part (2) of the counted proof (lines 901-910). Produces the explicit presentation exhibiting finite presentation.

Lemma 0.1. *Let $\varphi : R \to S$ be a ring map.*

(1) *If φ is finite, then φ is of finite type.*
(2) *If S is of finite presentation as an R-module, then φ is of finite presentation.*

Proof. For (1) if $x_1, \ldots, x_n \in S$ generate S as an R-module, then $x_1, \ldots, x_n$ generate S as an R-algebra. For (2), suppose that $\sum r_j^i x_i = 0$, $j = 1, \ldots, m$ is a set of generators of the relations among the x_i when viewed as R-module generators of S. Furthermore, write $1 = \sum r_i x_i$ for some $r_i \in R$ and $x_i x_j = \sum r_{ij}^k x_k$ for some $r_{ij}^k \in R$. Then

$$S = R[t_1, \ldots, t_n]/(\sum r_j^i t_i,\ 1 - \sum r_i t_i,\ t_i t_j - \sum r_{ij}^k t_k)$$

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as an R-algebra which proves (2). $\square$